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Title of Article

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Key Points:

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- RSTF

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- Source Inversion using RSTF and Teleseismic data

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- Compare slip models from RSTF and Teleseismic data

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Abstract

Will be done Later.

Plain Language Summary

Will be done Later.

1 Introduction

Earthquakes represent one of the most significant natural hazards...

1.1 Elastic Rebound Theory and Earthquake Cycles on Transform Faults

- Application of elastic rebound theory and seismic gap hypotheses to stress accumulation and release along fault boundaries.
- Shortcomings of the elastic rebound theory in explaining complex earthquake cycles, especially on oceanic transform faults.

1.2 Seismotectonic Setting of our study area and the fault structure

- Tectonic framework of the plate boundary, quasi-periodic seismicity, fault strike, and plate motion. (strike/dip/rake, slip rate, etc.)
- Structural characteristics of the fault plane: non-vertical, complex geometry.
- Overview of the 2015 and 2025 earthquake sequence occurring along the same fault system within a 10-year span. (figure on the seismicity and fault structure)

1.3 Unresolved Questions and Study Objectives

- In this slowly moving transform fault system, the 2015 and 2025 earthquakes occurred within a short time span of 10 years which is unexpected. This contradicts the natural hypothesis of stress loading and release along the fault system.
- Are the slip distributions of the 2015 and 2025 earthquakes similar or different? If different, what are the implications for stress accumulation and release along the fault system?

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2 Data and Methodology**Open Research Section**

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: [https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors](https://www.agu.org/Publish-with-AGU/Publish/Author-Resources/Data-for-Authors)

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Please include a comprehensive conflict of interest statement that reflect all conflicts of interest for all involved authors. If there are no conflicts of interest please state, “The authors declare there are no conflicts of interest for this manuscript.”

Acknowledgments

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